

Quality-aware wireless urban sensor networks using deep-learning

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Abstract

The proliferation of large-scale wireless sensor networks in smart cities generates vast streams of real-time data, but inherent quality issues present significant challenges for predictive analytics, particularly for complex systems like urban mobility. This paper investigates the relationship between data quality dimensions and the performance of deep learning models designed to predict pedestrian flows using data from the UK's Newcastle Urban Observatory. We develop a quality-aware machine learning pipeline, comparing univariate and multivariate Long Short-Term Memory (LSTM) models to forecast pedestrian activity. Our analysis reveals a critical insight for big data applications: the length of consecutive data sequences is a more significant determinant of model performance than overall data completeness. Specifically, sensors with high data completeness but with average sequence lengths below 20 hours produced higher prediction errors, a result attributed to the failure to capture crucial daily periodicities in movement patterns. Furthermore, we demonstrate that for longer forecasting horizons, multivariate models that encode periodic features can enhance prediction accuracy over univariate approaches, helping to mitigate the impact of data gaps. These findings establish key data quality requirements for reliably forecasting urban dynamics and provide a framework for developing the robust, quality-aware predictive systems necessary for real-time decision-making in big data-driven smart city applications.

Keywords: Wireless Sensor Networks, Pedestrian Monitoring, Data Quality, Machine Learning, Movement Prediction

List of Abbreviations

Abbreviation	Term
IoT	Internet of Things
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
MAPE	Mean Absolute Percentage Error
UV	Univariate
MV	Multivariate

1 Introduction

The Internet of Things (IoT) and wireless sensor networks have become fundamental components of smart city infrastructure, enabling real-time monitoring and analysis of urban environments [1–3]. Wireless sensor networks consist of spatially distributed autonomous sensors that cooperatively monitor physical or environmental conditions, transmitting data wirelessly to central gateways for processing and analysis [4]. Data quality in wireless sensor networks is critically important for making informed decisions about urban dynamics [5]. Current cities typically have limited decision-making functionality due to slow, incomplete, and largely disconnected information transmission systems [6]. However, as cities deploy more extensive networks of low-cost sensors to improve spatial resolution, they face increased data quality challenges [7]. Low-cost sensors are more prone to errors and inconsistencies than their expensive counterparts, and the volume of data generated makes manual quality assurance infeasible [8]. This necessitates automated systems that are ‘quality-aware’ and capable of assessing and managing data quality in real-time [9, 10].

While significant research exists on data quality for wireless sensor networks in environmental monitoring applications [10–12], there remains a distinct gap in understanding data quality requirements for complex dynamic systems such as urban

mobility [13]. Urban mobility represents a complex adaptive system of systems, characterised by intricate interactions between pedestrian flows, transportation networks, and urban infrastructure [14]. The intellectual challenge lies in accurately predicting the spatio-temporal dynamics of these interconnected systems in real-time - a task made particularly difficult by the non-linear nature of human movement patterns and the inherent variability in sensor data quality [5, 7]. This predictive capability is a key enabling technology for urban digital twins, which promise to revolutionise city management through real-time emergency response, infrastructure maintenance optimisation, and resource monitoring [15, 16].

The critical prerequisite for reliable prediction is understanding the degree to which the sensor network data can be trusted, requiring real-time 'quality awareness' of incoming data streams [8, 17]. This paper begins to address this gap by investigating the relationship between data quality dimensions (different measures of data quality such as completeness) and the performance of a model predicting complex spatio-temporal dynamics of pedestrian movements [8].

2 Literature Review

Current approaches to wireless sensor network data quality management reveal significant limitations when applied to urban mobility prediction. Traditional frameworks, such as those proposed by Wang et al. [18] and later refined by Karkouch et al. [7], emphasise static quality dimensions like accuracy and completeness but fail to adequately address the temporal dependencies crucial for movement prediction. While these frameworks provide valuable taxonomies for categorising data quality issues, they offer limited practical guidance for handling the dynamic nature of urban mobility data.

The limitations of existing approaches become particularly evident in real-time applications. Current quality control mechanisms, as reviewed by Teh et al. [8], typically rely on post hoc detection and correction of data quality issues, making them unsuitable for real-time decision support systems. Furthermore, most existing studies have focused on synthetic datasets or controlled environments [19, 20], leaving a significant gap in understanding how these approaches perform with the messy, incomplete data streams characteristic of real-world urban sensor networks. This gap is particularly problematic for pedestrian monitoring systems, where data quality issues can arise from various sources including environmental conditions, sensor malfunctions, and communication failures [17].

Teh et al. finds that the most common approaches to error detection and fault correction use artificial neural networks [8]. Aouedi et al. find that traditional methods such autoregressive integrated moving average (ARIMA) struggle to model the non-linearity and complex dependencies often found in traffic data, leading to lower prediction accuracy, especially for longer-term forecasting [21]. Liu et al. compare performance of deep learning models for traffic forecasting, including LSTMs and transformers [22]. They find that while transformers achieved higher prediction accuracy, they generally have a larger number of parameters (millions) compared to LSTMs (hundreds of thousands), stating that transformer-based models can "suffer from long training and inference times".

3 Methodology

3.1 Data Collection

The [Newcastle Urban Observatory](#), one of the largest public-facing urban sensing networks in the United Kingdom, provides real-time monitoring of various urban metrics across Newcastle upon Tyne. Each monitoring station integrates three key components: an IP-camera for video capture, a Raspberry Pi processing unit enhanced

with a Google Coral TPU AI-accelerator, and a cellular modem for data transmission [23]. The system conducts real-time video analytics, performing object detection and tracking while recording line-crossing events. This processed data is then transmitted to a central backend system, which serves as our primary data source for subsequent analysis.

The data from the pedestrian sensors is aggregated at 15-minute intervals, with each record containing the number of pedestrians detected within a scene over that time period. Figure 1 A shows the raw data from a single sensor over 18 days, highlighting two distinct patterns of missing data: regular nighttime gaps (a design feature of some sensors) and extended periods of data sparsity due to technical issues such as sensor failure or data transmission problems. Figure 1 B quantifies this data incompleteness, there should ideally be 96 records per day (4 per hour), few days achieve complete coverage and some days have no records at all.

3.2 Data Preprocessing

The data preprocessing stage executes functions related to data cleaning (data removal, data interpolation, and data sequencing), that are necessary before any feature engineering takes place. Whilst there are deep learning models that can handle missing data, these models tend to be more complex and tuning them can be challenging [24]. Another option is using complex imputation methods to fill in the missing data, but these are often costly and can affect the integrity of the data [24]. For these reasons, the data has been left in its raw form, and a consecutive sequence detection and labelling algorithm has been developed [25]. Consecutive sequences are sections of the time series that exist without any missing data. It works by scanning through a DataFrame row by row, building sequences where the time difference between consecutive rows equals the recording interval of the sensor (normally 15-minutes). When there's a gap in the timestamps or when, it checks if the current sequence meets the

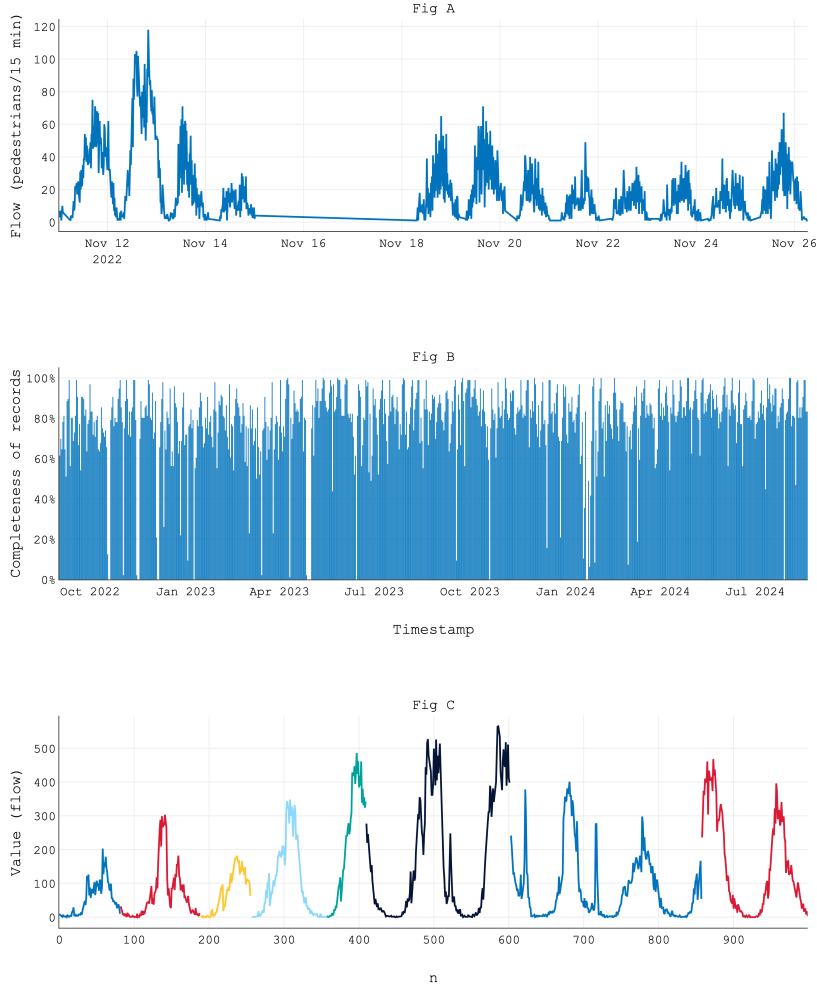


Fig. 1 Raw data (A), percentage of records recorded per day (B), and preprocessed data for an example sensor (C).

minimum length requirement ($\text{window_size} + \text{horizon}$ - see Figure 2 for a visualisation of the parameters). Valid sequences are stored and numbered sequentially.

Two key hyper-parameters define the window structure [26]:

1. *window_size*: Determines how many historical time steps are included in each window. This parameter can be tuned to optimise model performance.

2. *horizon*: Specifies how far into the future the model predicts. This parameter is not tuned and is set based on the desired prediction length.

Figure 1 C shows the outputs of the data preprocessing stage for the same sensor as A and B where n is the temporal index of each observation and *value* is the number of pedestrians observed. Each new consecutive sequence is shown as a new colour. The sequences in this example capture similar segments of the daily cycle before a missing record appears. This means each sequence is less than 96 records long which limits the length of training windows that can be used in the machine learning pipeline (for example a window size of 24 with a horizon of 24 requires at least 48 consecutive time steps, otherwise the sequence will be discarded).

3.3 Feature Engineering

The feature engineering stage tests the hypothesis that incorporating explicit temporal features alongside raw sensor data could improve prediction accuracy, particularly for longer horizons. This hypothesis proposes that combining harmonic decomposition through Lomb-Scargle periodograms [27, 28] with deep learning would enable better capture of both cyclical patterns and irregular fluctuations in pedestrian movement. The proposed dual representation strategy - using both raw time series and engineered temporal features - was designed to provide a more robust foundation for handling the inherent periodicities in urban mobility data while maintaining resilience to gaps in sensor observations. To test this hypothesis, the methodology specifically differentiates between short-term (2-hour) and longer-term (6-hour) prediction horizons to evaluate how feature complexity requirements scale with prediction length.

The features are then normalised using the Standard Scaler equation $X'_i = \frac{X_i - \mu}{\sigma}$, where X_i is the original value, μ is the mean of the feature, and σ is the standard deviation of the feature. The standard scalar normalisation technique is chosen to ensure that any future data points falling outside of the existing range can be scaled

323 appropriately as a min max scaler would not be appropriate in this case [29]. Calculating the mean and standard deviation in the scaling process also serves as a check for future data drift; if movement patterns change over the time the mean is likely to be non-stationary.

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331 3.4 Data Loading

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333 For recurrent neural networks (such as LSTMs), a number of approaches to data loading can be used. A common approach for time-series data is using fixed-sized windows [30, 31]. An example fixed-size window is shown in Figure 2 (n is the temporal index of each observation and *value* is the scaled pedestrian count). A fixed-size window contains multiple sequences of data: the primary sensor readings (shown in blue) and additional engineering features (shown in grey). Each window is associated with a label (shown in red) which represents a future data point that the model aims to predict.

346 In the example plot, a window size of 24 and horizon of 24 are used. This means the model is trained to predict pedestrian flow 6 hours into the future using the previous 6 hours of data. The prediction is made recursively, using the standardised count values and the additional temporal features from within each window.

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354 3.5 Model Training

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356 Two primary experiments were conducted to evaluate model performance: an interpolation experiment and a multivariate/univariate (MV/UV) comparison. The interpolation experiment examined whether filling nighttime data gaps (between 00:00 and 06:00) with zero values would improve model performance. The multivariate/univariate comparison assessed whether incorporating additional engineered features, such as temporal patterns and frequency components, would enhance prediction accuracy

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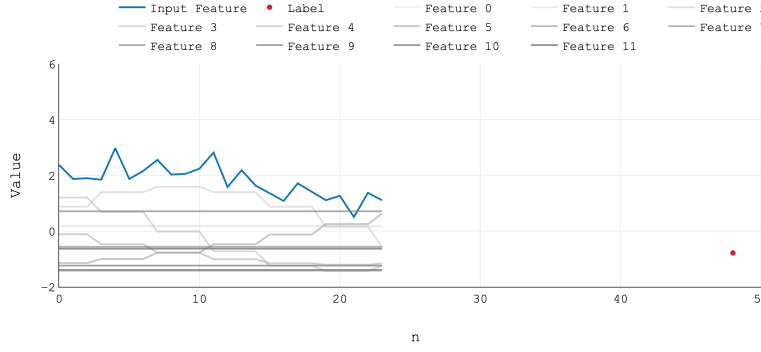


Fig. 2 Example window used in the data loading process.

compared to using only raw pedestrian counts. Both experiments measured comparative performance across twelve sensors selected for their superior data completeness (75%-85%).

For both experiments, we systematically tuned the models' hyper-parameters through an optimisation process consisting of 100 trials per experimental condition. The hyper-parameters and their possible values are presented in Table 1. These parameters include the window size (ranging from 4 to 48 time steps), batch size (32 to 128), learning rate (10^{-5} to 10^{-1}), scheduler step size (3 to 7 steps), model architecture (LSTM or GRU), hidden dimensions (32 to 256 units), number of layers (1 to 3), and dropout rate (0 to 0.5). While each experimental condition resulted in different optimal hyper-parameter values, we maintained consistent parameter ranges across all conditions to ensure a fair comparison.

To tune the models hyper-parameters, the Optuna library [32] is used in conjunction with MLflow [33] for experiment tracking. The process involves running the pipeline multiple times with different hyper-parameters and recording the results. RMSE (Root Mean Square Error) was selected as the optimisation metric for hyper-parameter tuning, following established practices in time-series prediction tasks [34].

3.6 Model Evaluation

Table 1 Hyper-parameter tuning configuration

Parameter	Suggestion Type	Value Range/Options
window_size	suggest_categorical	[4, 8, 12, 16, 24, 32, 48]
batch_size	suggest_categorical	[32, 64, 128]
lr	suggest_loguniform	1e-5 to 1e-1
scheduler_step_size	suggest_int	3-7
model_type	suggest_categorical	["lstm", "gru"]
hidden_dim	suggest_categorical	[32, 64, 128, 256]
num_layers	suggest_int	1 to 3
dropout	suggest_uniform	0.0 to 0.5

To evaluate the model a range of metrics are used. The primary metric (used for hyperparameter tuning) will be the root mean squared error (RMSE). RMSE measures the difference between the predicted and actual values and is more sensitive to outliers than the mean absolute error (MAE). Mean absolute percentage error (MAPE) will be used as it can show the relative error between predicted and actual values. R-squared (R^2) metric is also used. R^2 measures the proportion of the variance in the dependent variable that is predictable from the independent variable(s). The R^2 scores are expected to be low due to amount of noise in the data.

Sensor data quality is evaluated using two primary metrics: average sequence length and data completeness for both experiments. Average sequence length, defined as the length of consecutive record sequences, provides insight into data continuity. Data completeness is calculated as the ratio of actual records to expected records throughout the study period.

3.7 Hyper-parameter Tuning

The hyperparameter optimisation, detailed in Table 2, reveal distinct architectural differences among models. The UV-8 model emerged as the most compact, utilising

Table 2 Comparison of tuned hyper-parameters for different experiments. The window size was consistently selected as 24 time steps (6 hours). All models selected the same batch size. Hidden dimensions and number of layers vary between experiments. The smallest model is UV8 and the largest is UV24.

	UV-24	MV-24	UV-8	MV-8
window_size	24	24	24	24
horizon	24	24	8	8
stride	1	1	1	1
batch_size	32	32	32	32
model_type	lstm	lstm	lstm	lstm
hidden_dim	256	64	64	64
num_layers	2	2	1	2
dropout	0.064	0.053	0.072	0.173
learning_rate	0.00011	0.00024	0.00382	0.00012
scheduler_step_size	7	4	4	3
scheduler_gamma	0.518	0.564	0.440	0.802

a single layer with 64 hidden dimensions. In contrast, MV models maintained consistent architectures with 64 hidden dimensions across two layers. The UV-24 model proved to be the largest, implementing 256 hidden dimensions across two layers. This configuration may raise concerns about potential over-parameterisation, as the UV-24 model contains four times the hidden units of the MV-24 model despite processing fewer input features (1 versus 12). Additionally, the MV-8 model’s elevated dropout rate of 17.3% (compared to 5-7% in other models) suggests a higher susceptibility to overfitting, necessitating more aggressive regularisation.

4 Results

4.1 Univariate / Multivariate Experiment

Tables 3 and 4 show the results for the 12 most complete sensors for the multivariate/univariate experiment. Where MV performs better than UV, the value is highlighted in yellow, and where UV performs better than MV the value is highlighted in teal. Analysis of these results show that for the smaller prediction horizon (2 hours), the

univariate models generally outperformed the multivariate models across the 3 metrics of MAPE, RMSE and MAE. For the longer prediction horizon (6-hours) the multivariate models achieved marginally better performance metrics, exhibiting lower MAE and RMSE than their univariate counterparts (although the differences are very small). The difference in MAPE is still in favour of the univariate models. This pattern could suggest either an insufficient capture of temporal dependencies in the univariate models or a reliance on periodicity features for prediction. The strong performance of the univariate approach for the 2-hour models may be attributed to the effectiveness of recent historical values as predictors, whereas introducing multivariate features may decrease the signal to noise ratio.

Table 3 Comparison of test metrics for MV and UV models with a horizon of 8 timesteps (2 hours).

UV	Test MAPE			UV	Test RMSE			UV	Test MAE		
	MV	diff			MV	diff			MV	diff	
2.148	3.308	1.160		0.623	0.656	0.033		0.484	0.515	0.031	
1.632	1.922	0.290		0.752	0.683	-0.069		0.604	0.542	-0.063	
1.755	2.047	0.292		0.701	0.680	-0.021		0.528	0.515	-0.013	
1.650	2.317	0.667		0.632	0.683	0.051		0.485	0.549	0.064	
1.332	1.509	0.177		0.639	0.639	0.000		0.478	0.474	-0.003	
16.165	13.558	-2.607		0.384	0.387	0.003		0.282	0.283	0.002	
1.123	1.174	0.051		0.380	0.373	-0.007		0.270	0.275	0.005	
2.221	2.504	0.283		0.405	0.464	0.059		0.304	0.366	0.062	
0.865	0.755	-0.110		0.414	0.397	-0.017		0.297	0.310	0.012	
1.466	1.869	0.403		0.603	0.614	0.011		0.458	0.479	0.021	
1.446	2.289	0.843		0.528	0.605	0.077		0.407	0.476	0.068	
1.986	2.212	0.226		0.748	0.681	-0.067		0.600	0.531	-0.069	

Examining the relationship between the data quality metrics of completeness and sequence length help to establish a relation between model performance and the quality of the training data from the sensor. Figure 3 presents the relationship between Root Mean Square Error (RMSE) and these metrics, respectively. Each bar corresponds to a sensor, ordered by decreasing RMSE. The completeness values for these sensors vary between 0.75 and 0.85 and the RMSE values are between 0.35 and 1.0.

Table 4 Comparison of test metrics for MV and UV models with a horizon of 24 timesteps (6-hours).

Test MAPE			Test RMSE			Test MAE		
UV	MV	diff	UV	MV	diff	UV	MV	diff
1.623	1.945	0.322	0.696	0.671	-0.025	0.546	0.535	-0.011
2.790	3.648	0.858	0.928	0.830	-0.098	0.760	0.680	-0.081
2.463	2.687	0.224	0.680	0.675	-0.005	0.509	0.520	0.011
1.431	1.442	0.011	0.665	0.610	-0.055	0.506	0.465	-0.040
15.748	15.547	-0.201	0.629	0.602	-0.027	0.470	0.448	-0.021
1.653	1.270	-0.383	0.492	0.442	-0.050	0.369	0.371	0.001
1.324	1.180	-0.144	0.450	0.356	-0.094	0.315	0.271	-0.044
1.321	0.980	-0.341	0.533	0.431	-0.102	0.400	0.359	-0.041
0.827	0.930	0.103	0.437	0.356	-0.081	0.313	0.277	-0.035
17.913	22.985	5.072	0.593	0.594	0.001	0.465	0.479	0.015
1.646	2.990	1.344	0.513	0.667	0.154	0.390	0.547	0.156
3.008	4.392	1.384	0.861	0.848	-0.013	0.681	0.668	-0.013

From the graph it appears that sensors with less data (lower completeness) have lower RMSE. Whilst this appears counterintuitive - one would expect that more data would lead to better model performance. Average sequence length provides important context that explains this pattern. The average sequence lengths vary from 80 to 150 and the RMSE between 0.35 and 1.0. The sensors with the highest RMSE have the shortest average sequence length of around 20 hours which are also the sensors with the highest data completeness. The lowest RMSE values are observed for sensors with an average sequence of around 35 hours. The distinct difference in RMSE values between sensors with different average sequence lengths could be due to the model's inability to capture the main temporal dependencies in the data. The most powerful signal from the Lomb-Scargle frequency analysis occurs at around 24 hours which corresponds to daily patterns of in pedestrians activity (morning and evening commutes, lunchtime etc.). It seems plausible that the sensors with <20 hours average sequence length are not capturing the main periodicity in the data as effectively, leading to higher RMSE values, despite having more records to train on.

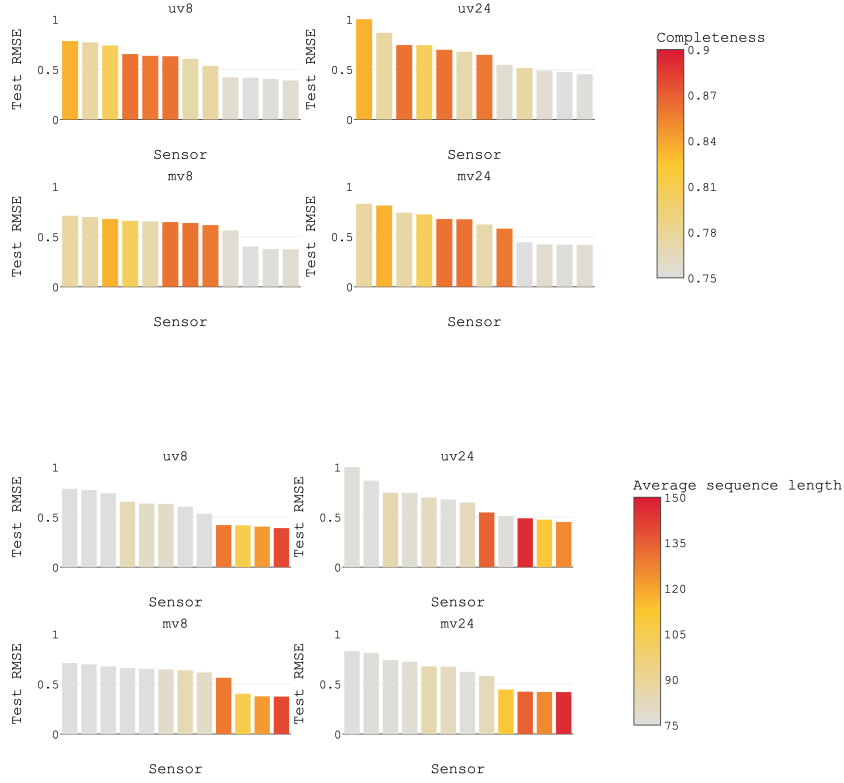


Fig. 3 RSME vs data completeness and average sequence length for the 12 sensors with the most complete data.

4.2 Interpolation Experiment

For the interpolation experiment, missing values between 00:00 and 06:00 were replaced with zeros. These specific hours were chosen based on historical pedestrian activity in the city centre, which typically show minimal footfall traffic during these early morning hours, making zero values a reasonable approximation for most locations. As many of the sensors were programmed to not record when zero pedestrians were detected, the idea was to see if the short sequence lengths associated with these sensors could be remedied. Figure 4 illustrates the resultant effect on average sequence

length, demonstrating a substantial increase from 20–40 hours to 80–200 hours. However, as evidenced in Table 5, this interpolation yields negligible improvements in model RMSE. The limited effectiveness can be attributed to the simplistic nature of the interpolation method, which assumes zero pedestrian activity during nighttime hours across all sensors. This assumption proves particularly problematic for sensors that experience significant nighttime activity *and* extensive missing periods, as it introduces artificial patterns of zero values into the training data during extended missing periods. A potential refinement would involve expanding the training window from 24 time steps (6 hours) to 48 time steps (12 hours), thereby enabling the model to better contextualise isolated patterns of zero values.

4.3 Predictive Modelling Performance

Following these experimental outcomes, the feasibility of predicting model performance using the data quality metrics was investigated. The ability to predict model performance from data quality metrics would provide valuable guidance for sensor network deployments, helping organisations assess whether their data collection methods are likely to yield reliable predictive models before investing significant resources in model development. A Random Forest regressor with K-fold cross-validation to predict the Mean Absolute Error (MAE) of the LSTM models was used with predictor variables, identified through feature importance analysis [35, 36], comprising of average sequence length, data completeness, sequence count, and the standard deviation of sequence lengths. Figure 5 presents the results of this analysis. It was found that the standard deviation of sequence length was the most important predictor, however when using this feature alone the model struggles to predict performance, indicating the importance of using a variety of metrics for estimating data quality. While the model achieved an R-squared value of 0.35, indicating modest correlation, this performance metric should be interpreted cautiously due to potential overfitting on the

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Table 5 Comparison between raw and interpolated data for **multivariate** models with a horizon of 24 hours.

Sensor Name	Test MAE		Test Std.	
	Raw	Int	Raw	Int
GREYSTHEATRESOUTH_NORTHWEST_TO_SOUTHWEST	0.547	0.584	0.533	0.656
GREYSTHEATRESOUTH_SOUTHWEST_TO_NORTHWEST	0.479	0.597	0.609	0.760
NCLNORTHUMBERLANDSTSAVILLEROWEAST_NORTHEAST_TO_SOUTHEAST	0.277	0.336	0.390	0.513
NCLNORTHUMBERLANDSTSAVILLEROWEAST_SOUTHEAST_TO_NORTHEAST	0.359	0.359	0.514	0.555
NCLNORTHUMBERLANDSTSAVILLEROWEST_NORTHWEST_TO_SOUTHWEST	0.271	0.324	0.400	0.495
NCLNORTHUMBERLANDSTSAVILLEROWEST_SOUTHWEST_TO_NORTHWEST	0.371	0.336	0.514	0.513
NCLPILGRIMSTMARKETLN_FROM_NORTH_TO_SOUTH	0.448	0.466	0.650	0.675
NCLPILGRIMSTMARKETLN_FROM_SOUTH_TO_NORTH	0.465	0.500	0.702	0.721
NCLSIDESTCROWNPOSADANORTH_FROM_EAST_TO_WEST	0.668	0.886	0.951	0.904
NCLSIDESTCROWNPOSADANORTH_FROM_WEST_TO_EAST	0.520	0.612	0.822	0.745
NCLSIDESTCROWNPOSADASOUTH_FROM_EAST_TO_WEST	0.680	0.558	0.905	0.752
NCLSIDESTCROWNPOSADASOUTH_FROM_WEST_TO_EAST	0.535	0.496	0.752	0.649

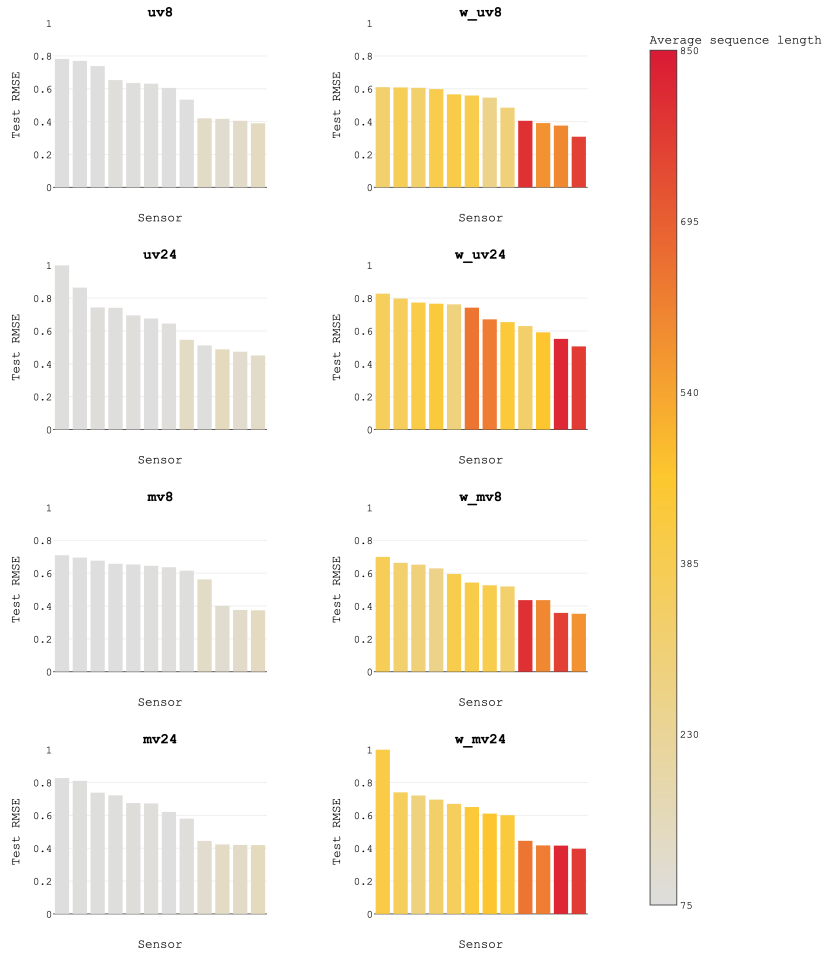


Fig. 4 Comparison of average sequence length after the preprocessing stage for each of the 8 variations.

limited sensor dataset. The four features all show relatively high importance in predicting the MAE of the LSTM models. Figure 5 also show the effect of each feature on the predicted MAE.

It may be that using R^2 for correlation is not the best metric for this analysis, as the relationship between the data quality metrics and the model performance is not necessarily linear. The partial dependence plots in Figure 5 show that the relationship

783 between the data quality metrics and the model performance is not linear. For exam-
784 ple, the average sequence length has a positive effect on the model performance up to
785 a certain point, after which the effect is negative. This suggests that the relationship
786 between the data quality metrics and the model performance is more complex than a
787 simple linear relationship.
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793 **5 Discussion**

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796 **5.1 Data Quality Assessment and Implications**

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798 The investigation into IoT pedestrian data streams reveals that data quality challenges
799 stem from multiple factors, aligning with [7]’s comprehensive analysis of IoT data
800 quality issues. While their work established a broad framework for understanding IoT
801 data quality, our findings specifically demonstrate the critical importance of temporal
802 continuity in data sequences for predictive modelling performance. The relationship
803 between sequence length and model performance, particularly the threshold of 20
804 hours for capturing daily periodicities, addresses a gap in existing IoT data quality
805 frameworks. This finding extends the traditional data quality taxonomies by emphasising
806 the temporal dimension of data quality in urban sensing applications. Our analysis
807 revealed that sensors with shorter average sequence lengths (<20 hours) exhibited
808 poorer predictive performance despite higher overall data completeness rates. Sensors
809 with >80% completeness achieved RMSE values >0.5 whereas the best performing
810 sensors had 75% completeness. The same sensors with lowest completeness had the
811 longest average sequence lengths (around 35 hrs for the 75% complete sensors, compared
812 with 20 hrs for the >80% complete sensors). This apparently counterintuitive
813 result can be explained through the lens of [8]’s work on temporal dependencies in
814 sensor networks. The inability to capture complete diurnal cycles significantly impairs
815 model performance, suggesting that traditional data quality metrics may need revision
816 for time-series applications in urban environments.
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Fig. 5 MV24 Random Forest partial dependence plots and feature importance.

The comparative performance of univariate and multivariate models provides insight into the relationship between data quality and model complexity. Our findings align with [37]’s observations regarding the importance of feature engineering in time-series analysis, while extending their work to specifically address urban pedestrian

875 monitoring applications. The marginally better performance of multivariate models
876 for longer prediction horizons, particularly in lower quality datasets, suggests addi-
877 tional features that capture some of the higher order temporal dependencies may help
878 compensate for data quality issues. It is however unclear whether it is worthwhile
881 including these multivariate features given the computational overhead of including
882 engineered features in the training process. Further research will need to be done on
883 larger prediction horizons to determine whether there is a trend between perfor-
886 mance, prediction horizon and the number of features included in the model. The
887 average sequence lengths in the sensor data were insufficient to conduct these exper-
888 iments using the existing fixed time window methodology. It seems plausible that for
891 much longer prediction horizons (up to 24 hours) that the multivariate models would
893 significantly outperform the univariate models making the feature engineering process
894 worthwhile.

896 Attempts at simple interpolation proved largely ineffective, supporting [24]’s
897 assertion that complex imputation methods might be necessary for meaningful
898 improvement. Future work will need to investigate whether sophisticated interpolation
901 can rival the performance of the sensors that capture full diurnal cycles.

903 There is some evidence to suggest that simple measures of data quality could be
904 used to predict whether the data in a network of sensors is of sufficient quality to be
905 used for predictive modelling for different prediction horizons. The framework that
906 has been presented as part of this research could be used by organisations to ensure
908 their sensor networks contain enough data to be used for predictive modelling, before
911 expensive and time-consuming machine learning pipelines are developed.

5.2 Implementation, Limitations, and Practical	921
Recommendations	922
The findings from this research have led to specific modifications in data collection	923
protocols for the Urban Observatory’s pedestrian sensors, particularly the implemen-	924
tation of completeness metadata and sensor heartbeat monitoring. Sensor heartbeat	925
monitoring outputs a boolean value that allows us to differentiate between periods	926
where the sensor is not outputting data because there is no activity and when the	927
sensor is not working. These changes align with [17]’s recommendations for enhanced	928
data provenance in IoT systems, while addressing the specific challenges of pedestrian	929
monitoring. The introduction of sensor heartbeat monitoring, in particular, represents	930
a practical application of [38]’s proactive data quality management principles. Our	931
quality-aware pipeline demonstrates the feasibility of automated quality assessment	932
in real-time sensor networks, addressing a key challenge identified by [9].	933
This study faced several methodological constraints that should be considered	934
when interpreting its findings. A primary limitation was the rapid drop-off in data	935
availability for prediction horizons beyond 6 hours, stemming from the fixed-window	936
training approach and nighttime data gaps. The requirement for consecutive data	937
windows meant that even a 6-hour prediction horizon needed at least 12-16 hours of	938
continuous data when accounting for context and overlap. The approach of optimising	939
for average performance across all sensors, while computationally efficient, potentially	940
undermined opportunities for sensor-specific optimisation and may have resulted in	941
overtraining for sensors with lower data completeness. Additionally, while our random	942
forest model showed modest success ($R^2 = 0.35$) in predicting deep-learning model	943
performance based on data quality metrics, these results should be interpreted cau-	944
tiously given the limited sensor dataset. The absence of cross-fold validation means	945
the model lacks exposure to the most recent 15% of data, potentially limiting its	946
adaptability to evolving urban mobility patterns.	947
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967 5.3 Summary

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969 This study has revealed several insights about the relationship between data quality
970 and predictive modelling performance in urban sensor networks. The analysis demon-
971 strates that above a certain threshold of completeness (around 75% of the number of
972 the total number of records) average sequence length plays an important role in model
973 training accuracy - particularly when the average sequence captures diurnal cycles.
974 The investigation into multivariate versus univariate approaches suggests potential
975 advantages of multivariate models for longer prediction horizons, though the bene-
976 fits must be weighed against increased computational overhead. Furthermore, while
977 simple interpolation techniques proved ineffective for improving model performance,
978 the study established that basic data quality metrics could serve as indicators for
979 determining whether sensor data is suitable for predictive modelling applications.

980
981 Future research should focus on addressing these limitations through several key
982 avenues. The development of sensor-specific optimisation strategies would allow for
983 more nuanced model tuning, potentially improving performance across diverse sen-
984 sor configurations. Methods that utilise spatial dependencies between sensors could
985 enhance prediction accuracy by leveraging the inherent relationships in pedestrian
986 movement patterns across urban spaces. Integration of sophisticated anomaly detec-
987 tion algorithms, possibly incorporating multiple data quality dimensions, would enable
988 more robust identification of sensor malfunctions and data quality issues, allowing
989 these data points to be removed from a dataset, enhancing the quality of the training
990 data. Finally, validation of these findings across larger sensor networks and diverse
991 urban environments would strengthen the generalisability of the results and pro-
992 vide insights into scaling these approaches for broader smart city applications. These
993 developments would further contribute to the robust and practical implementations
994 of quality-aware predictive modelling systems that are critical in enabling real-time
995 decision-making and management for urban environments.

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Declarations	1013
	1014
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not involve direct interaction with human participants.	1046
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Consent for publication Not applicable. The manuscript does not contain any	1049
individual person's data in any form.	1050
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Data availability The pedestrian count data analysed during the current study	1053
are publicly available from the Newcastle Urban Observatory portal [39].	1054
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Materials availability Not applicable.	1057
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1059 **Code availability** The code developed for the data preprocessing and sequencing
1060 described in this paper is available on GitHub [25].

1062 **Author contribution** CMW: Conceptualisation, Methodology, Software, Vali-
1063 dation, Formal analysis, Investigation, Data Curation, Writing, Visualisation. SB:
1064 Conceptualisation, Supervision – Review & Editing. PJ: Supervision – Review &
1065 Editing. TK: Software, Data Curation

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